

The Charge-Conservation Hinge

One conserved ledger, changing nuclei and particles

Working rule: In a closed system, $\sum Q_{\text{before}} = \sum Q_{\text{after}}$. In nuclide notation ${}^A_Z\text{X}$, Z is proton number and A is nucleon number. Charge and nucleon-number conservation constrain a nuclear equation; they do not determine whether a decay occurs, its half-life, or how the released energy is shared. In these nuclear equations, beta particles are written explicitly and neutrinos carry zero charge.

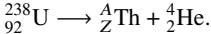
Name: _____

Class: _____

Companion task to [The Placeholder Force](#)

1. Alpha decay: a charged cluster leaves

Complete the alpha decay of uranium-238:



An alpha particle is a helium-4 nucleus containing two protons and two neutrons.

A. Balance the two ledgers

Use nucleon-number conservation to find A , and charge conservation to find Z . Write the completed equation. Display the two checks separately:

$$A: \quad 238 = \cdots, \quad Z: \quad 92 = \cdots.$$

B. Interpret the change

State how the parent nucleus changes in the (A, Z) description. Explain why the chemical element changes. On a small sketch of a nuclide chart, represent the alpha step as an arrow.

C. What does conservation determine?

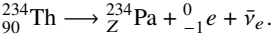
Charge and nucleon number fix the identity of the daughter once alpha emission is specified. List three things they do *not* determine, such as whether the decay is energetically allowed, its half-life, or the kinetic-energy division.

D. Supply the missing nuclear physics

Alpha decay requires a favourable mass-energy difference and quantum tunnelling through the Coulomb barrier. Explain why this is additional closure rather than another form of charge bookkeeping.

2. Beta-minus decay: charge is redistributed

Thorium-234 undergoes beta-minus decay. Complete



At particle level, the weak interaction converts a neutron according to $n \rightarrow p + e^- + \bar{\nu}_e$.

A. Balance the charge ledger

Find Z and write the completed equation. Show explicitly that

$$+90 = (+Z) + (-1) + 0.$$

Why is the nucleon number unchanged even though the element changes?

B. Follow the microscopic conversion

Use $n \rightarrow p + e^- + \bar{\nu}_e$ to explain why proton number increases by one. Clarify why the emitted electron should not be pictured as a pre-existing electron stored inside the nucleus.

C. Why include the antineutrino?

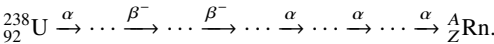
Electric charge conservation alone would permit the equation without a charged neutrino. Explain what the antineutrino contributes to the fuller account: energy and momentum sharing, angular momentum, and lepton-number conservation.

D. Construct the beta-plus analogue

A proton-rich nucleus undergoes beta-plus decay. Complete the generic changes $\Delta A =$ _____, $\Delta Z =$ _____, and write the particle conversion beginning with a proton. Check the charge ledger.

3. A decay cascade: many steps, one ledger

A section of the uranium-238 series contains four alpha decays and two beta-minus decays:



A. Build the cascade one step at a time

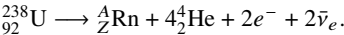
Complete the sequence:



Label each arrow α or β^- . Check A and Z after every step.

B. Compress the entire cascade

Treat all six decays as one aggregate equation:



Find A and Z . Verify both the total nucleon-number ledger and the total charge ledger.

C. Read the path as vector steps

An alpha step has $(\Delta A, \Delta Z) = (-4, -2)$; a beta-minus step has $(0, +1)$. Add four alpha vectors and two beta-minus vectors. Show that the resultant maps uranium-238 to the final radon nuclide.

D. What is absent from the ledger?

Explain why conservation alone does not determine the order of the six decays, the intermediate nuclides, the branching ratios, or the time spent at each stage.

Synthesis: conservation balances the nuclear equation; interaction physics selects the decay

1. Complete the comparison.

Process	Microscopic change	ΔA	ΔZ	Additional physics required
Alpha				
Beta minus				
Beta plus				

2. Develop the central claim.

Respond to:

“*Charge conservation tells us how to balance a decay equation. It does not tell us which decay nature will choose.*”

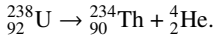
Use the terms *closed system*, *nuclear charge*, *nucleon number*, *alpha particle*, *beta particle*, *neutrino*, *weak interaction*, *binding energy*, *branching ratio*, and *half-life*. Explain why a balanced equation is necessary but insufficient as a physical theory of radioactive decay.

Teacher guide and indicative answers

The Charge-Conservation Hinge

1. Alpha decay: conservation constrains the daughter

Balanced equation.



The ledgers are $238 = 234 + 4$ and $92 = 90 + 2$. Thus $(\Delta A, \Delta Z) = (-4, -2)$ for the daughter nucleus.

Interpretation. Removing two protons changes uranium ($Z = 92$) to thorium ($Z = 90$). On a nuclide chart, alpha decay moves two places down in proton number and four down in mass number, equivalently two down in Z and two down in neutron number $N = A - Z$.

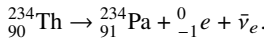
What conservation omits. It does not determine the Q -value, whether the parent state is unstable to alpha emission, the tunnelling probability, half-life, angular-momentum selection constraints, or the kinetic-energy division between daughter and alpha particle.

Closure. Nuclear masses and binding energies determine whether the process is energetically allowed. Barrier penetration supplies the decay probability. These are dynamical and quantum-mechanical statements, not bookkeeping identities.

Teaching caution. A balanced alpha equation identifies the only daughter compatible with the stated decay mode. It does not prove that the decay occurs.

2. Beta decay: a weak-interaction conversion

Balanced beta-minus equation.



Charge: $+90 = +91 - 1 + 0$. Nucleon number remains 234 because a neutron becomes a proton; the total number of nucleons is unchanged.

Particle process.

$$n \rightarrow p + e^- + \bar{\nu}_e.$$

The electron and antineutrino are produced in the weak interaction. The old picture of an electron already resident in the nucleus is misleading.

Why the antineutrino matters. It is electrically neutral, so charge conservation does not demand it. It is required in the full Standard Model account to conserve lepton number and to share energy, momentum and angular momentum, explaining the continuous beta spectrum.

Beta-plus analogue.

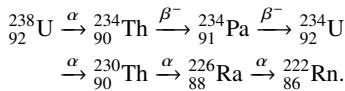
$$p \rightarrow n + e^+ + \nu_e, \quad (\Delta A, \Delta Z) = (0, -1).$$

Charge: $+1 = 0 + 1 + 0$. For a parent nucleus, Z decreases by one. Positron emission also requires sufficient mass-energy; electron capture can compete.

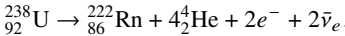
Teaching caution. Z labels proton number, not the net charge of the neutral atom. Be consistent about nuclear versus atomic masses when later calculating Q -values.

3. Cascade: local steps and a global ledger

Completed sequence.



Aggregate equation.



Nucleon number: $238 = 222 + 4(4)$. Charge: $92 = 86 + 4(2) + 2(-1) = 92$.

Vector sum.

$$4(-4, -2) + 2(0, +1) = (-16, -6),$$

so $(238, 92) \mapsto (222, 86)$.

What the aggregate hides. The global ledger cannot reconstruct the order of steps, the identities and lifetimes of intermediates, competing branches, decay energies, or selection rules. Those require the nuclear energy-level structure and the relevant strong, electromagnetic and weak-interaction dynamics.

Useful extension. Ask students whether a different ordering of four alpha and two beta-minus symbols would preserve the same final (A, Z) . It would, but most such paths would not correspond to physically available nuclides or allowed decays.

Teacher synthesis: a conserved quantity is a constraint, not a decay mechanism

Process	Microscopic change	ΔA	ΔZ	Additional physics
Alpha	emission of a $2p + 2n$ cluster	-4	-2	Q -value, Coulomb barrier, tunnelling
Beta minus	$n \rightarrow p + e^- + \bar{\nu}_e$	0	$+1$	weak interaction, lepton number, phase space
Beta plus	$p \rightarrow n + e^+ + \nu_e$	0	-1	weak interaction, energy threshold, competition with capture

A strong response should distinguish three levels:

Ledger: total electric charge and nucleon number balance across the chosen closed system.

Identity: once a decay mode is specified, the conservation equations constrain the daughter nuclide and emitted particles.

Dynamics: mass-energy differences, nuclear structure, interaction type and quantum transition probabilities determine whether the decay occurs, its products, branching ratio and half-life.

The central sentence is: *Conservation filters possible nuclear stories; it does not write the story.*